Solar Ti appendix

Sun · Ti · downloadable appendix

Document	Solar Ti appendix
Subject	Solar photospheric abundance appendix · Ti
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Solar titanium: 4.945 ± 0.065

Asplund, Amarsi & Grevesse 2021: 4.97 ± 0.05 · this measurement -0.025 dex

Asplund, Amarsi & Grevesse (2021) give no line recipe we can repeat for this element, so this is our best HARPS product: the Reference Grade analysis with the most complete treatment, then the most lines.

VIS: 4.945 ± 0.065 · HARPS · VIS · Ti I · reference lines · <3D> LTE (Gerber)

Highlighted Products

near-UV

5.106 ± 0.047

Ti II · reference lines · <3D> LTE (Gerber) · Kitt Peak (1984, molecfit) · Reference Grade

VIS

4.940 ± 0.062

Ti I · reference lines · <3D> LTE (Gerber) · Kitt Peak (Kurucz 2005) · Reference Grade

red-optical

5.113 ± 0.094

Ti I · reference lines · <3D> NLTE (Gerber) · Kitt Peak (Kurucz 2005) · Reference Grade

NIR

5.022 ± 0.172

Ti I · reference lines · <3D> LTE (Gerber) · Kitt Peak (1984, molecfit) · Reference Grade

H

4.864 ± 0.122

Ti I · reference lines · <3D> NLTE (Gerber) · CRIRES+ H (full arm, telluric-corrected) · Reference Grade

K

4.932 ± 0.158

Ti I · reference lines · <3D> NLTE (Gerber) · CRIRES+ · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	198 products -- Ti I · Asplund et al. 2021 lines; Ti I · reference lines; Ti II · Asplund et al. 2021 lines; Ti II · reference lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	144 products -- Ti I · Codex-graded lines; Ti II · Codex-graded lines
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

near-UV — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.68–5.62

Ti II · Codex-graded lines · 1D LTE Codex Grade · 5 lines		5.082 ±0.122
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 5 lines		5.082 ±0.123
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 4 lines		5.118 ±0.111
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 5 lines		5.067 ±0.119
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.107 ±0.101
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.202 ±0.113
Ti II · reference lines · 1D LTE Reference Grade · 153 lines		5.183 ±0.048
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 147 lines		5.184 ±0.049
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 96 lines		5.101 ±0.041
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 153 lines		5.203 ±0.048
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 154 lines · EXPERIMENTAL-NOT-ADOPTED		5.361 ±0.048
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 135 lines · EXPERIMENTAL-NOT-ADOPTED		5.289 ±0.043
Ti I · Codex-graded lines · 1D LTE Codex Grade · 15 lines		5.206 ±0.168
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 15 lines		5.248 ±0.167
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 8 lines		5.140 ±0.142
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 14 lines		5.204 ±0.151
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 14 lines · EXPERIMENTAL-NOT-ADOPTED		5.348 ±0.145

near-UV — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.68–5.62

Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 12 lines · EXPERIMENTAL-NOT-ADOPTED		5.455 ±0.123
Ti I · reference lines · 1D LTE Reference Grade · 34 lines		5.293 ±0.104
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 34 lines		5.318 ±0.104
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 15 lines		5.067 ±0.084
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 33 lines		5.303 ±0.101
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 31 lines · EXPERIMENTAL-NOT-ADOPTED		5.459 ±0.108
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 26 lines · EXPERIMENTAL-NOT-ADOPTED		5.506 ±0.084

near-UV — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.68–5.62

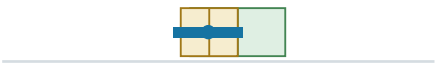
Ti II · Codex-graded lines · 1D LTE Codex Grade · 5 lines		4.845 ±0.125
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 5 lines		4.836 ±0.125
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		4.948 ±0.141
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 5 lines		4.854 ±0.117
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 5 lines · EXPERIMENTAL-NOT-ADOPTED		4.946 ±0.106
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 5 lines · EXPERIMENTAL-NOT-ADOPTED		5.035 ±0.124
Ti II · reference lines · 1D LTE Reference Grade · 153 lines		5.003 ±0.046
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 147 lines		5.012 ±0.047
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 121 lines		4.965 ±0.040
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 154 lines		5.016 ±0.047

near-UV — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.68–5.62

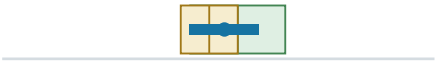
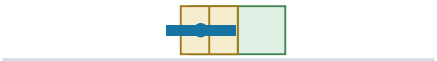
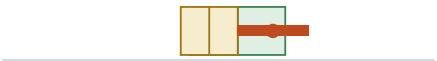
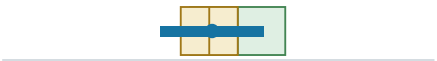
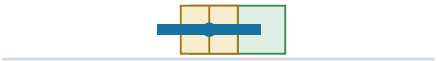
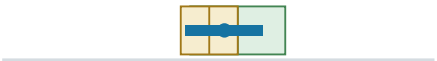
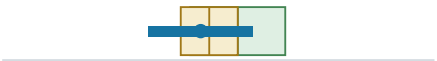
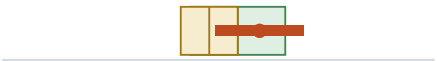
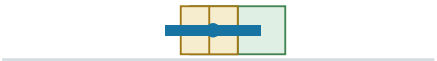
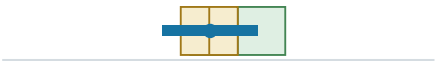
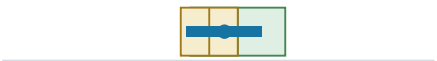
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 156 lines · EXPERIMENTAL-NOT-ADOPTED		5.106 ±0.047
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 142 lines · EXPERIMENTAL-NOT-ADOPTED		5.114 ±0.043
Ti I · Codex-graded lines · 1D LTE Codex Grade · 12 lines		5.047 ±0.129
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 12 lines		5.091 ±0.129
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 10 lines		5.067 ±0.136
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 12 lines		5.055 ±0.130
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 14 lines · EXPERIMENTAL-NOT-ADOPTED		5.186 ±0.141
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 13 lines · EXPERIMENTAL-NOT-ADOPTED		5.233 ±0.130
Ti I · reference lines · 1D LTE Reference Grade · 36 lines		4.963 ±0.092
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 36 lines		4.988 ±0.092
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 28 lines		5.074 ±0.074
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 34 lines		5.006 ±0.076
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 35 lines · EXPERIMENTAL-NOT-ADOPTED		5.164 ±0.084
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 35 lines · EXPERIMENTAL-NOT-ADOPTED		5.280 ±0.080

near-UV axis: 4.68 · 4.92 · 5.15 · 5.38 · 5.62 A(Ti) dex

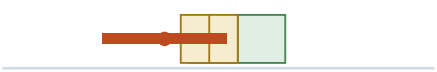
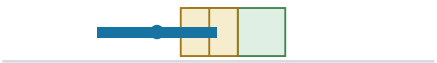
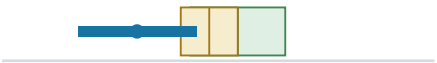
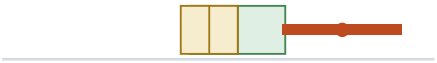
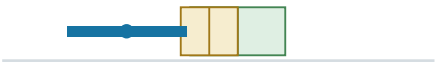
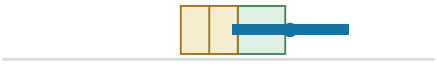
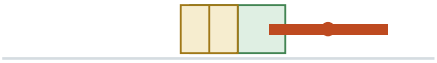
VIS — HARPS
A(Ti) dex; axis 4.72–5.18

Ti II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 13 lines		4.939 ±0.037
Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 13 lines		4.938 ±0.037

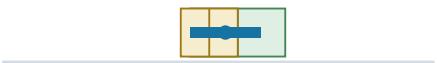
VIS — HARPS
A(Ti) dex; axis 4.72–5.18

Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 13 lines		4.956 ±0.037
Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 13 lines		4.931 ±0.036
Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 13 lines · EXPERIMENTAL-NOT-ADOPTED		4.993 ±0.038
Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 13 lines · EXPERIMENTAL-NOT-ADOPTED		5.007 ±0.037
Ti II · Codex-graded lines · 1D LTE Codex Grade · 49 lines		4.943 ±0.055
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 49 lines		4.940 ±0.055
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 47 lines		4.956 ±0.041
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 49 lines		4.931 ±0.055
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 48 lines · EXPERIMENTAL-NOT-ADOPTED		4.993 ±0.046
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 48 lines · EXPERIMENTAL-NOT-ADOPTED		5.014 ±0.044
Ti II · reference lines · 1D LTE Reference Grade · 72 lines		4.944 ±0.050
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 72 lines		4.941 ±0.050
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 70 lines		4.956 ±0.040
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 72 lines		4.932 ±0.051
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 72 lines · EXPERIMENTAL-NOT-ADOPTED		4.995 ±0.043
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 71 lines · EXPERIMENTAL-NOT-ADOPTED		5.023 ±0.042
Ti I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 27 lines		4.819 ±0.067
Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 26 lines		4.845 ±0.068

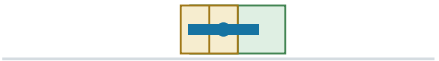
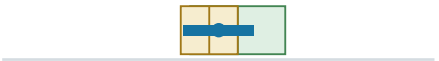
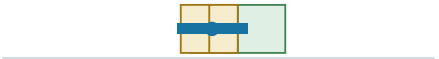
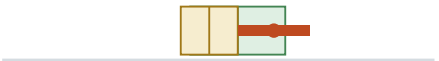
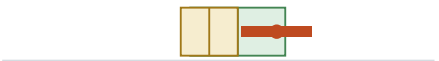
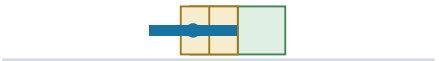
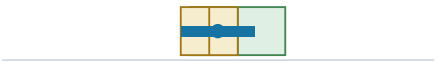
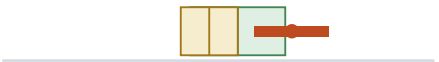
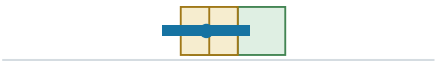
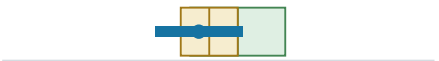
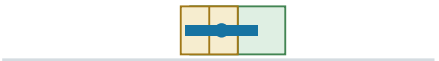
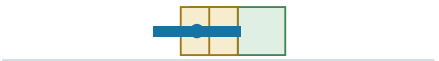
VIS — HARPS
A(Ti) dex; axis 4.72–5.18

Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 27 lines		5.027 ±0.065
Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 27 lines		4.832 ±0.067
Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 27 lines · EXPERIMENTAL-NOT-ADOPTED		4.893 ±0.065
Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 27 lines · EXPERIMENTAL-NOT-ADOPTED		5.053 ±0.063
Ti I · Codex-graded lines · 1D LTE Codex Grade · 99 lines		4.848 ±0.063
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 97 lines		4.885 ±0.063
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 90 lines		5.028 ±0.062
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 98 lines		4.864 ±0.063
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 104 lines · EXPERIMENTAL-NOT-ADOPTED		4.942 ±0.064
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 100 lines · EXPERIMENTAL-NOT-ADOPTED		5.080 ±0.063
Ti I · reference lines · 1D LTE Reference Grade · 77 lines		4.838 ±0.063
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 77 lines		4.853 ±0.063
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 73 lines		5.025 ±0.061
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 77 lines		4.844 ±0.063
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 80 lines · EXPERIMENTAL-NOT-ADOPTED		4.945 ±0.065
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 77 lines · EXPERIMENTAL-NOT-ADOPTED		5.065 ±0.062

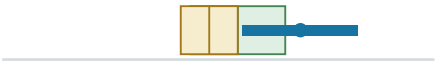
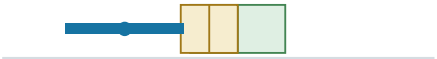
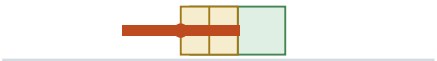
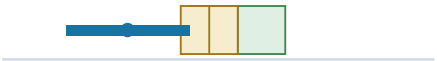
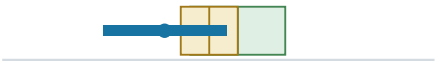
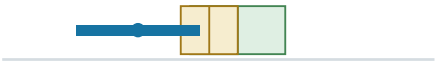
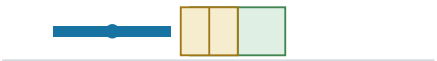
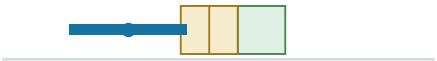
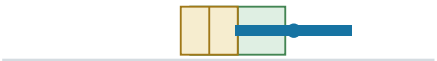
VIS — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.72–5.18

Ti II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 14 lines		4.957 ±0.037

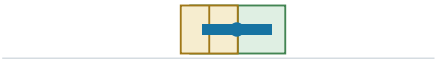
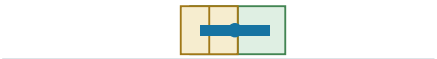
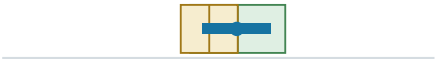
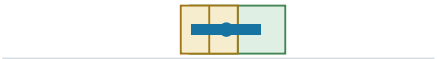
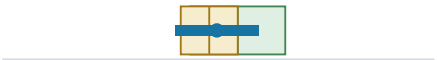
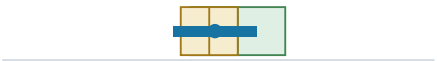
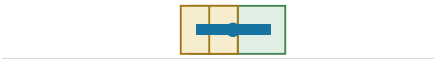
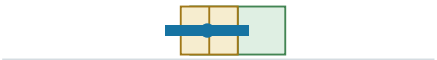
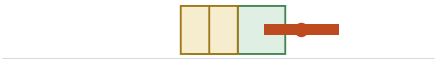
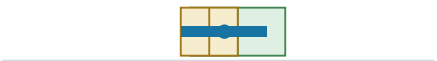
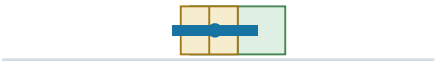
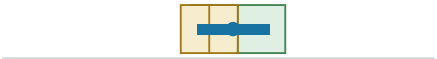
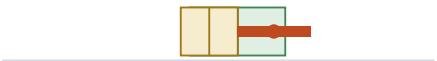
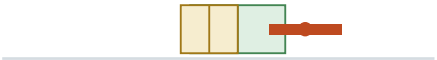
VIS — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.72–5.18

Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 14 lines		4.955 ±0.037
Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 14 lines		4.950 ±0.037
Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 14 lines		4.943 ±0.037
Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 14 lines · EXPERIMENTAL-NOT-ADOPTED		5.008 ±0.037
Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 14 lines · EXPERIMENTAL-NOT-ADOPTED		5.011 ±0.037
Ti II · Codex-graded lines · 1D LTE Codex Grade · 49 lines		4.926 ±0.046
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 49 lines		4.923 ±0.046
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 48 lines		4.949 ±0.039
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 49 lines		4.917 ±0.046
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 48 lines · EXPERIMENTAL-NOT-ADOPTED		4.990 ±0.040
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 48 lines · EXPERIMENTAL-NOT-ADOPTED		5.027 ±0.039
Ti II · reference lines · 1D LTE Reference Grade · 73 lines		4.937 ±0.046
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 73 lines		4.929 ±0.046
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 71 lines		4.953 ±0.038
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 73 lines		4.927 ±0.046
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 73 lines · EXPERIMENTAL-NOT-ADOPTED		4.996 ±0.039
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 72 lines · EXPERIMENTAL-NOT-ADOPTED		5.030 ±0.038
Ti I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 27 lines		4.839 ±0.062

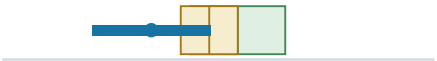
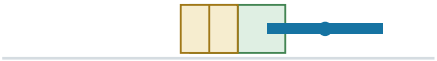
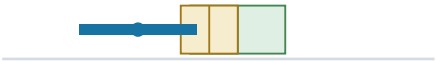
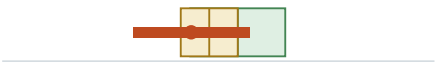
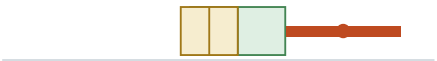
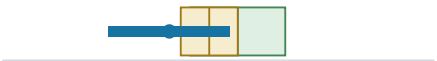
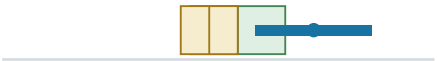
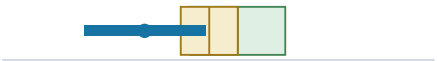
VIS — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.72–5.18

Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 26 lines		4.869 ±0.063
Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 27 lines		5.036 ±0.061
Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 27 lines		4.851 ±0.062
Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 27 lines · EXPERIMENTAL-NOT-ADOPTED		4.910 ±0.062
Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 27 lines · EXPERIMENTAL-NOT-ADOPTED		5.074 ±0.060
Ti I · Codex-graded lines · 1D LTE Codex Grade · 104 lines		4.854 ±0.065
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 102 lines		4.893 ±0.065
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 97 lines		5.043 ±0.061
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 104 lines		4.865 ±0.065
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 106 lines · EXPERIMENTAL-NOT-ADOPTED		4.947 ±0.063
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 102 lines · EXPERIMENTAL-NOT-ADOPTED		5.085 ±0.062
Ti I · reference lines · 1D LTE Reference Grade · 78 lines		4.838 ±0.062
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 78 lines		4.855 ±0.062
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 76 lines		5.029 ±0.061
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 78 lines		4.854 ±0.062
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 80 lines · EXPERIMENTAL-NOT-ADOPTED		4.940 ±0.062
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 78 lines · EXPERIMENTAL-NOT-ADOPTED		5.058 ±0.062

VIS — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.72–5.18

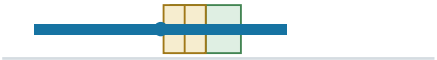
Ti II · Asplund et al. 2021 lines · 1D LTE Reference Grade · 14 lines		4.969 ±0.037
Ti II · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 14 lines		4.967 ±0.037
Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 14 lines		4.969 ±0.036
Ti II · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 14 lines		4.958 ±0.036
Ti II · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 14 lines · EXPERIMENTAL-NOT-ADOPTED		5.024 ±0.037
Ti II · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 14 lines · EXPERIMENTAL-NOT-ADOPTED		5.031 ±0.036
Ti II · Codex-graded lines · 1D LTE Codex Grade · 49 lines		4.948 ±0.044
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 49 lines		4.946 ±0.044
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 48 lines		4.965 ±0.039
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 49 lines		4.938 ±0.044
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 48 lines · EXPERIMENTAL-NOT-ADOPTED		4.995 ±0.039
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 48 lines · EXPERIMENTAL-NOT-ADOPTED		5.037 ±0.039
Ti II · reference lines · 1D LTE Reference Grade · 73 lines		4.956 ±0.045
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 73 lines		4.946 ±0.045
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 71 lines		4.965 ±0.038
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 73 lines		4.945 ±0.045
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 73 lines · EXPERIMENTAL-NOT-ADOPTED		5.008 ±0.039
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 72 lines · EXPERIMENTAL-NOT-ADOPTED		5.041 ±0.038

VIS — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.72–5.18

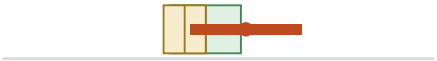
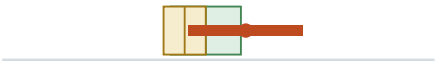
Ti I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 27 lines		4.851 ±0.062
Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 26 lines		4.879 ±0.062
Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 27 lines		5.062 ±0.061
Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 27 lines		4.865 ±0.062
Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 27 lines · EXPERIMENTAL-NOT-ADOPTED		4.921 ±0.062
Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 27 lines · EXPERIMENTAL-NOT-ADOPTED		5.081 ±0.060
Ti I · Codex-graded lines · 1D LTE Codex Grade · 102 lines		4.859 ±0.064
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 100 lines		4.898 ±0.064
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 96 lines		5.050 ±0.061
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 102 lines		4.872 ±0.064
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 105 lines · EXPERIMENTAL-NOT-ADOPTED		4.951 ±0.063
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 101 lines · EXPERIMENTAL-NOT-ADOPTED		5.085 ±0.061
Ti I · reference lines · 1D LTE Reference Grade · 78 lines		4.853 ±0.061
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 78 lines		4.866 ±0.061
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 75 lines		5.036 ±0.061
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 78 lines		4.865 ±0.061
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 80 lines · EXPERIMENTAL-NOT-ADOPTED		4.939 ±0.062
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 78 lines · EXPERIMENTAL-NOT-ADOPTED		5.075 ±0.062

VIS axis: 4.72 · 4.84 · 4.95 · 5.06 · 5.18 A(Ti) dex

red-optical — IAG FTS
A(Ti) dex; axis 4.68–5.30

Ti II · Codex-graded lines · 1D LTE Codex Grade · 6 lines		5.018 ±0.124
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 6 lines		5.016 ±0.126
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		4.958 ±0.091
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 6 lines		5.006 ±0.121
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.053 ±0.137
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.048 ±0.131
Ti II · reference lines · 1D LTE Reference Grade · 6 lines		5.018 ±0.124
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 6 lines		5.016 ±0.126
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 5 lines		4.958 ±0.091
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 6 lines		5.006 ±0.121
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.053 ±0.137
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.048 ±0.131
Ti I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		4.894 ±0.180
Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		4.919 ±0.173
Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 7 lines		5.109 ±0.156
Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 7 lines		4.906 ±0.180
Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 7 lines · EXPERIMENTAL-NOT-ADOPTED		4.976 ±0.154
Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 7 lines · EXPERIMENTAL-NOT-ADOPTED		5.126 ±0.132

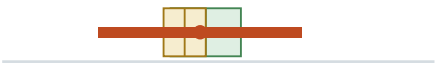
red-optical — IAG FTS
A(Ti) dex; axis 4.68–5.30

Ti I · Codex-graded lines · 1D LTE Codex Grade · 30 lines		4.918 ±0.082
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 29 lines		4.922 ±0.083
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 28 lines		5.094 ±0.077
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 30 lines		4.931 ±0.082
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 30 lines · EXPERIMENTAL-NOT-ADOPTED		5.027 ±0.080
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 30 lines · EXPERIMENTAL-NOT-ADOPTED		5.133 ±0.077
Ti I · reference lines · 1D LTE Reference Grade · 28 lines		4.918 ±0.084
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 27 lines		4.922 ±0.085
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 26 lines		5.094 ±0.079
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 28 lines		4.931 ±0.084
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 28 lines · EXPERIMENTAL-NOT-ADOPTED		5.027 ±0.081
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 28 lines · EXPERIMENTAL-NOT-ADOPTED		5.133 ±0.079

red-optical — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.68–5.30

Ti II · Codex-graded lines · 1D LTE Codex Grade · 6 lines		5.025 ±0.113
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 6 lines		5.023 ±0.114
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		4.958 ±0.085
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 6 lines		5.013 ±0.109
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.059 ±0.125

red-optical — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.68–5.30

Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.059 ±0.121
Ti II · reference lines · 1D LTE Reference Grade · 6 lines		5.025 ±0.113
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 6 lines		5.023 ±0.114
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 5 lines		4.958 ±0.085
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 6 lines		5.013 ±0.109
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.059 ±0.125
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.059 ±0.121
Ti I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		4.879 ±0.169
Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		4.904 ±0.162
Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 7 lines		5.080 ±0.139
Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 7 lines		4.892 ±0.165
Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 7 lines · EXPERIMENTAL-NOT-ADOPTED		4.962 ±0.145
Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 7 lines · EXPERIMENTAL-NOT-ADOPTED		5.108 ±0.126
Ti I · Codex-graded lines · 1D LTE Codex Grade · 31 lines		4.917 ±0.086
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 30 lines		4.909 ±0.087
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 29 lines		5.072 ±0.081
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 31 lines		4.929 ±0.084
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 32 lines · EXPERIMENTAL-NOT-ADOPTED		5.012 ±0.092

red-optical — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.68–5.30

Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 32 lines · EXPERIMENTAL-NOT-ADOPTED		5.120 ±0.090
Ti I · reference lines · 1D LTE Reference Grade · 29 lines		4.915 ±0.089
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 28 lines		4.907 ±0.090
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 27 lines		5.071 ±0.083
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 29 lines		4.927 ±0.087
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 30 lines · EXPERIMENTAL-NOT-ADOPTED		5.012 ±0.095
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 30 lines · EXPERIMENTAL-NOT-ADOPTED		5.113 ±0.094

red-optical — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.68–5.30

Ti II · Codex-graded lines · 1D LTE Codex Grade · 6 lines		5.034 ±0.117
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 6 lines		5.032 ±0.118
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		4.986 ±0.089
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 6 lines		5.023 ±0.114
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.069 ±0.128
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.069 ±0.124
Ti II · reference lines · 1D LTE Reference Grade · 6 lines		5.034 ±0.117
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 6 lines		5.032 ±0.118
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 5 lines		4.986 ±0.089
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 6 lines		5.023 ±0.114

red-optical — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.68–5.30

Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.069 ±0.128
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.069 ±0.124
Ti I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 7 lines		4.888 ±0.154
Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		4.913 ±0.148
Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) Reference Grade · 7 lines		5.111 ±0.130
Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) Reference Grade · 7 lines		4.900 ±0.152
Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) Reference Grade · 7 lines · EXPERIMENTAL-NOT-ADOPTED		4.970 ±0.136
Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) Reference Grade · 7 lines · EXPERIMENTAL-NOT-ADOPTED		5.124 ±0.118
Ti I · Codex-graded lines · 1D LTE Codex Grade · 30 lines		4.912 ±0.081
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 29 lines		4.913 ±0.082
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 28 lines		5.085 ±0.075
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 30 lines		4.925 ±0.080
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 30 lines · EXPERIMENTAL-NOT-ADOPTED		5.012 ±0.079
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 30 lines · EXPERIMENTAL-NOT-ADOPTED		5.127 ±0.077
Ti I · reference lines · 1D LTE Reference Grade · 28 lines		4.912 ±0.083
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 27 lines		4.913 ±0.084
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 26 lines		5.085 ±0.077
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 28 lines		4.925 ±0.081

red-optical — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.68–5.30

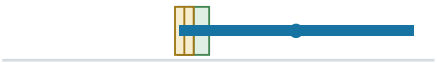
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 28 lines · EXPERIMENTAL-NOT-ADOPTED		5.012 ±0.079
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 28 lines · EXPERIMENTAL-NOT-ADOPTED		5.127 ±0.078

red-optical axis: 4.68 · 4.83 · 4.99 · 5.14 · 5.30 A(Ti) dex

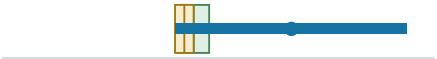
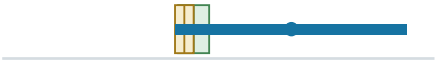
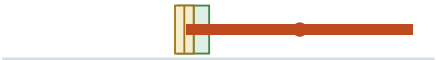
NIR — IAG FTS
A(Ti) dex; axis 4.35–5.75

Ti II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		4.808 ±0.084
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		4.801 ±0.083
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		4.806 ±0.069
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		4.796 ±0.083
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		4.832 ±0.089
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		4.833 ±0.079
Ti II · reference lines · 1D LTE Reference Grade · 2 lines		4.808 ±0.084
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		4.801 ±0.083
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 2 lines		4.806 ±0.069
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 2 lines		4.796 ±0.083
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		4.832 ±0.089
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		4.833 ±0.079

NIR — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.35–5.75

Ti II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		5.300 ±0.377

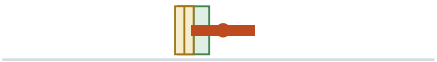
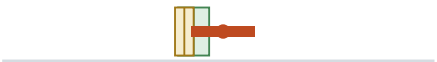
NIR — Kitt Peak (Kurucz 2005)
A(Ti) dex; axis 4.35–5.75

Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		5.294 ±0.377
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 1 line		4.926 ±0.170
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		5.285 ±0.373
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		5.330 ±0.386
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		5.312 ±0.364
Ti II · reference lines · 1D LTE Reference Grade · 2 lines		5.300 ±0.377
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		5.294 ±0.377
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 1 line		4.926 ±0.170
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 2 lines		5.285 ±0.373
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		5.330 ±0.386
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		5.312 ±0.364

NIR — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.35–5.75

Ti II · Codex-graded lines · 1D LTE Codex Grade · 2 lines		4.841 ±0.041
Ti II · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		4.834 ±0.041
Ti II · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 2 lines		4.838 ±0.045
Ti II · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 2 lines		4.829 ±0.041
Ti II · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		4.868 ±0.037
Ti II · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		4.870 ±0.038

NIR — Kitt Peak (1984, molecfit)
A(Ti) dex; axis 4.35–5.75

Ti II · reference lines · 1D LTE Reference Grade · 2 lines		4.841 ±0.041
Ti II · reference lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		4.834 ±0.041
Ti II · reference lines · 1D NLTE (Gerber) Reference Grade · 2 lines		4.838 ±0.045
Ti II · reference lines · 1D LTE (Gerber) Reference Grade · 2 lines		4.829 ±0.041
Ti II · reference lines · <3D> LTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		4.868 ±0.037
Ti II · reference lines · <3D> NLTE (Gerber) Reference Grade · 2 lines · EXPERIMENTAL-NOT-ADOPTED		4.870 ±0.038
Ti I · Codex-graded lines · 1D LTE Codex Grade · 7 lines		4.927 ±0.176
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 7 lines		4.917 ±0.178
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 6 lines		5.006 ±0.099
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 7 lines		4.916 ±0.177
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 7 lines · EXPERIMENTAL-NOT-ADOPTED		5.022 ±0.172
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.065 ±0.102
Ti I · reference lines · 1D LTE Reference Grade · 7 lines		4.927 ±0.176
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 7 lines		4.917 ±0.178
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 6 lines		5.006 ±0.099
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 7 lines		4.916 ±0.177
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 7 lines · EXPERIMENTAL-NOT-ADOPTED		5.022 ±0.172
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		5.065 ±0.102

NIR — CRIRES+ J
A(Ti) dex; axis 4.35–5.75

Ti I · Codex-graded lines · 1D LTE Codex Grade · 6 lines		4.595 ±0.203
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 6 lines		4.587 ±0.204
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		4.798 ±0.137
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 6 lines		4.614 ±0.227
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		4.722 ±0.176
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		4.840 ±0.166
Ti I · reference lines · 1D LTE Reference Grade · 6 lines		4.595 ±0.203
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 6 lines		4.587 ±0.204
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 5 lines		4.798 ±0.137
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 6 lines		4.614 ±0.227
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		4.722 ±0.176
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 6 lines · EXPERIMENTAL-NOT-ADOPTED		4.840 ±0.166

NIR axis: 4.35 · 4.70 · 5.05 · 5.40 · 5.75 A(Ti) dex

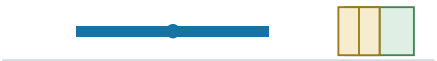
H — CRIRES+ H
A(Ti) dex; axis 4.42–5.05

Ti I · Codex-graded lines · 1D LTE Codex Grade · 5 lines		4.691 ±0.241
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 5 lines		4.718 ±0.241
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 4 lines		4.860 ±0.076
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 5 lines		4.705 ±0.240
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 5 lines · EXPERIMENTAL-NOT-ADOPTED		4.766 ±0.239

H — CRIRES+ H
A(Ti) dex; axis 4.42–5.05

Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 4 lines · EXPERIMENTAL-NOT-ADOPTED		4.875 ±0.080
Ti I · reference lines · 1D LTE Reference Grade · 5 lines		4.691 ±0.241
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 5 lines		4.718 ±0.241
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 4 lines		4.860 ±0.076
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 5 lines		4.705 ±0.240
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 5 lines · EXPERIMENTAL-NOT-ADOPTED		4.766 ±0.239
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 4 lines · EXPERIMENTAL-NOT-ADOPTED		4.875 ±0.080

H — CRIRES+ H (full arm, telluric-corrected)
A(Ti) dex; axis 4.42–5.05

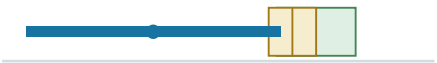
Ti I · Codex-graded lines · 1D LTE Codex Grade · 5 lines		4.669 ±0.140
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 5 lines		4.699 ±0.140
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 5 lines		4.866 ±0.134
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 5 lines		4.691 ±0.139
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 5 lines · EXPERIMENTAL-NOT-ADOPTED		4.749 ±0.128
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 5 lines · EXPERIMENTAL-NOT-ADOPTED		4.864 ±0.122
Ti I · reference lines · 1D LTE Reference Grade · 5 lines		4.669 ±0.140
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 5 lines		4.699 ±0.140
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 5 lines		4.866 ±0.134
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 5 lines		4.691 ±0.139

H — CRIRES+ H (full arm, telluric-corrected)
A(Ti) dex; axis 4.42–5.05

Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 5 lines · EXPERIMENTAL-NOT-ADOPTED		4.749 ±0.128
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 5 lines · EXPERIMENTAL-NOT-ADOPTED		4.864 ±0.122

H axis: 4.42 · 4.58 · 4.74 · 4.89 · 5.05 A(Ti) dex

K — CRIRES+
A(Ti) dex; axis 4.57–5.12

Ti I · Codex-graded lines · 1D LTE Codex Grade · 11 lines		4.748 ±0.142
Ti I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 11 lines		4.768 ±0.142
Ti I · Codex-graded lines · 1D NLTE (Gerber) Codex Grade · 9 lines		4.922 ±0.080
Ti I · Codex-graded lines · 1D LTE (Gerber) Codex Grade · 11 lines		4.764 ±0.161
Ti I · Codex-graded lines · <3D> LTE (Gerber) Codex Grade · 10 lines · EXPERIMENTAL-NOT-ADOPTED		4.853 ±0.125
Ti I · Codex-graded lines · <3D> NLTE (Gerber) Codex Grade · 11 lines · EXPERIMENTAL-NOT-ADOPTED		4.932 ±0.158
Ti I · reference lines · 1D LTE Reference Grade · 11 lines		4.748 ±0.142
Ti I · reference lines · 1D NLTE (Amarsi) Reference Grade · 11 lines		4.768 ±0.142
Ti I · reference lines · 1D NLTE (Gerber) Reference Grade · 9 lines		4.922 ±0.080
Ti I · reference lines · 1D LTE (Gerber) Reference Grade · 11 lines		4.764 ±0.161
Ti I · reference lines · <3D> LTE (Gerber) Reference Grade · 10 lines · EXPERIMENTAL-NOT-ADOPTED		4.853 ±0.125
Ti I · reference lines · <3D> NLTE (Gerber) Reference Grade · 11 lines · EXPERIMENTAL-NOT-ADOPTED		4.932 ±0.158

K axis: 4.57 · 4.71 · 4.85 · 4.98 · 5.12 A(Ti) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). For a molecule the NLTE term is replaced by its dependence on the other element, and the 3D term comes from the 3D-to-1D difference Amarsi et al. 2021 publish for the same band. Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the

supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every titanium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the titanium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	near-UV
What was measured	Ti I · Codex-graded lines; Ti II · Codex-graded lines
Solar spectra	Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	4–15
Result A(Ti)	4.946–5.348 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.13, model atmosphere 0.05

Band	near-UV
What was measured	Ti I · reference lines; Ti II · reference lines
Solar spectra	Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	15–156
Result A(Ti)	5.106–5.459 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.03, model atmosphere 0.02

Band	VIS
What was measured	Ti I · Codex-graded lines; Ti II · Codex-graded lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	47–106
Result A(Ti)	4.942–4.995 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	model atmosphere 0.05, fit / line-to-line scatter 0.03

Band	VIS
What was measured	Ti I · reference lines; Ti II · reference lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	70–80
Result A(Ti)	4.939–5.008 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	model atmosphere 0.05, fit / line-to-line scatter 0.02

Band	VIS
What was measured	Ti I · Asplund et al. 2021 lines; Ti II · Asplund et al. 2021 lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	13–27
Result A(Ti)	5.007–5.081 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	model atmosphere 0.05, fit / line-to-line scatter 0.02

Band	red-optical
What was measured	Ti I · Codex-graded lines; Ti II · Codex-graded lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	5–32
Result A(Ti)	5.048–5.133 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.06, model atmosphere 0.06

Band	red-optical
What was measured	Ti I · reference lines; Ti II · reference lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	5–30
Result A(Ti)	5.048–5.133 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.07, model atmosphere 0.06

Band	red-optical
What was measured	Ti I · Asplund et al. 2021 lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	7
Result A(Ti)	5.108–5.126 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.10, model atmosphere 0.06

Band	NIR
What was measured	Ti I · Codex-graded lines; Ti II · Codex-graded lines
Solar spectra	CRIRES+ J, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	1–7
Result A(Ti)	4.722–5.330 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.16, model atmosphere 0.05

Band	NIR
What was measured	Ti I · reference lines; Ti II · reference lines
Solar spectra	CRIRES+ J, IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	1–7
Result A(Ti)	4.722–5.330 (synth-mean3D-LTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.16, model atmosphere 0.05

Band	H
What was measured	Ti I · Codex-graded lines
Solar spectra	CRIRES+ H, CRIRES+ H (full arm, telluric-corrected)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	4–5
Result A(Ti)	4.864–4.875 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.11, model atmosphere 0.05

Band	H
What was measured	Ti I · reference lines
Solar spectra	CRIRES+ H, CRIRES+ H (full arm, telluric-corrected)
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	4–5
Result A(Ti)	4.864–4.875 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.11, model atmosphere 0.05

Band	K
What was measured	Ti I · Codex-graded lines
Solar spectra	CRIRES+
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	9–11
Result A(Ti)	4.932 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.15, model atmosphere 0.05

Band	K
What was measured	Ti I · reference lines
Solar spectra	CRIRES+
Treatments	1D-LTE, 1D-NLTE, synth-mean3D-LTE-gerber-stagger, synth-mean3D-NLTE-gerber-stagger
Lines	9–11
Result A(Ti)	4.932 (synth-mean3D-NLTE-gerber-stagger)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.15, model atmosphere 0.05

Solar titanium context

Titanium is measured from neutral Ti I and ionized Ti II lines from the near-ultraviolet to the near-infrared. The reference lines are the Ti I and Ti II lines Asplund et al. 2021 used, with the oscillator strengths they adopted; a wider set of lines with laboratory oscillator strengths and the Codex's own line selection are shown as separate products. Ti I departs from LTE, so each spectrum also carries non-LTE products. No titanium line in the H or K band has a published oscillator-strength uncertainty, so those bands are gaps.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Ti appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 4cbdd50cb115757aeacaa83122f1ae28489868b8
Feed Ti.json v1.342 · updated 2026-11-06T15:00:11Z
Generated 2026-11-06T18:07:59-07:00 (source commit timestamp)
Feed SHA-256 56996f31d23041cce17005e9346d92057e45d7a67eee8de9d95cb2b07448038c
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Ti comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. Ti II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Mallinson et al. 2024 departure grid via PySME (data/nlte_grids/Ti_Mallinson2024_PySME.csv). Sources: [5].

Model and grid sources. Feed treatment: ENGINE-B-NLTE; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Model and grid sources. Feed treatment: synth-1D-LTE-gerber; Turbospectrum on the registered Gerber deck. The product retains its own atmosphere and ion identity. Sources: [4].

Model and grid sources. Feed treatment: synth-mean3D-LTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [4], [6].

Model and grid sources. Feed treatment: synth-mean3D-NLTE-gerber-stagger; Turbospectrum on the registered Gerber deck with the horizontally averaged 3D (<3D>) Stagger solar model; not a full 3D calculation. Sources: [4], [6].

Solar abundance and indicator analyses. Published line set TI_AGSS21 (the line selection). Sources: [7].

Atomic data and gf authorities. Ti I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [9].

Product-to-source index

NIR · IAG FTS · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-7c65d6ac7e1e

NIR · IAG FTS · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-44b0ca4d0c54

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VIS · Kitt Peak (1984, molecfit) · Ti II · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [3], [4] Ti-products-a942d756e62c

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VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [8], [5] Ti-products-cd06ba9cfd18

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [8], [4] Ti-products-f521cfd6c101

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [8], [4] Ti-products-5c721eed5ca3

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [8], [4], [6] Ti-products-30af6935dca1

VIS · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [8], [4], [6] Ti-products-3cb64e50dabc

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-a7d756f143f7

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-7b4be941c8c1

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-9c061994e041

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-374d7ab89262

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-f7a046325126

VIS · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-2e8e5de89d78

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-c43d190e2acc

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-4aaf11a66098

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-4460d017cdb1

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-d9351369a0b6

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-509178a4cf19

near-UV · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-f1c1e95d6a37

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-206290b43cd9

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-63849d06ded0

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-f5ccf0395c54

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-121a17805bf9

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-6d1d43f16fe6

near-UV · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-dc897ab0bfe2

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-5e4b5154abd4

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-d87830f69068

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-d8ebd19befae

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-c7a18d1655b5

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-b4ca3df6eede

near-UV · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-71f5bd6dbabf

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-ef1616f06cbe

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-f88608578841

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-04880367c1d0

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-85cda524f86d

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-fb7cf7b619f1

near-UV · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-a52c3ee2aa12

red-optical · IAG FTS · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-181fd3ccbbd7

red-optical · IAG FTS · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-c7b92de79ee8

red-optical · IAG FTS · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-8aa32169d3dd

red-optical · IAG FTS · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-a1c67a8393aa

red-optical · IAG FTS · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-0481a85a39ff

red-optical · IAG FTS · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-cd6301a7115d

red-optical · IAG FTS · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-b9fd0a6e8b87

red-optical · IAG FTS · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-214266ebf9ba

red-optical · IAG FTS · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-23cf54ee763e

red-optical · IAG FTS · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-d305fb48b011

red-optical · IAG FTS · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-c722f2cf5fe5

red-optical · IAG FTS · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-5c3f6efd4d7b

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-7bd50ca0f43d

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-c78798edcaaf

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-231aab32af56

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-460068377c46

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-2f243605026b

red-optical · Kitt Peak (Kurucz 2005) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-b076a0a690d1

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-cac08c4eeac4

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-bef62d63ebec

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-4141f2b8fd61

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-d4c6e74deb11

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-70c2532d6eeb

red-optical · Kitt Peak (Kurucz 2005) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-64b3a157e60d

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Ti-products-8fbbb99e290e

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-d953f437f0fb

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-bf40832e7648

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-464acb032126

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-0045d6767454

red-optical · Kitt Peak (1984, molecfit) · Ti II · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-9c52791bd422

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE · [1], [2], [3], [4] Ti-products-5adbd47debf1

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Ti-products-3ac134114773

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D NLTE (Gerber) · [1], [2], [3], [4] Ti-products-dea62bec5093

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · 1D LTE (Gerber) · [1], [2], [3], [4] Ti-products-eefa1c832dd9

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> LTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-b0ce36bb68dd

red-optical · Kitt Peak (1984, molecfit) · Ti II · reference lines · <3D> NLTE (Gerber) · [1], [2], [3], [4], [6] Ti-products-4908df1ce17b

H · CRIRES+ H · Ti I · Codex-graded lines · 1D LTE · [1], [2], [9], [4] Ti-products-d2c4c83304df

H · CRIRES+ H · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [5] Ti-products-cb365f06d0e2

H · CRIRES+ H · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [4] Ti-products-61bd7949541d

H · CRIRES+ H · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [4] Ti-products-17f09adf6748

H · CRIRES+ H · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-fc90adf35fbf

H · CRIRES+ H · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-ecac625732a6

H · CRIRES+ H · Ti I · reference lines · 1D LTE · [1], [2], [9], [4] Ti-products-13a119d84a9d

H · CRIRES+ H · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [9], [5] Ti-products-f8ad6f5935b3

H · CRIRES+ H · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [4] Ti-products-83323fe9a34d

H · CRIRES+ H · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [4] Ti-products-de39a18fea2f

H · CRIRES+ H · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-bda553230b63

H · CRIRES+ H · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-dcb02f103251

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [9], [4] Ti-products-7693f3614ffe

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [5] Ti-products-8a951c905f50

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [4] Ti-products-f9d6cc25b73e

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [4] Ti-products-c0af0b068f2a

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-239e66a1258c

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-a446aa8db338

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · 1D LTE · [1], [2], [9], [4] Ti-products-ff9f10197c80

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [9], [5] Ti-products-3897f91261ac

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [4] Ti-products-393e2d7fa1f0

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [4] Ti-products-9d602c73d0ed

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-7d72f05fb19c

H · CRIRES+ H (full arm, telluric-corrected) · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [4], [6] Ti-products-60d0b83f1efa

K · CRIRES+ · Ti I · Codex-graded lines · 1D LTE · [1], [2], [9], [10], [4] Ti-products-6afd09595f1d

K · CRIRES+ · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10], [5] Ti-products-4e199512b892

K · CRIRES+ · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-3c767896ee6f

K · CRIRES+ · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-cae64fae86f1

K · CRIRES+ · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-69348f48b35c

K · CRIRES+ · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-1d74911ffc4e

K · CRIRES+ · Ti I · reference lines · 1D LTE · [1], [2], [9], [10], [4] Ti-products-07d5c23f8a30

K · CRIRES+ · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [9], [10], [5] Ti-products-581f743fb43d

K · CRIRES+ · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-3318fe09823e

K · CRIRES+ · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-80b36661ce64

K · CRIRES+ · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-48b486e5d256

K · CRIRES+ · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-c6ecb50573ad

NIR · CRIRES+ J · Ti I · Codex-graded lines · 1D LTE · [1], [2], [10], [4] Ti-products-c8e71b8ad992

NIR · CRIRES+ J · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [10], [5] Ti-products-6d015675e3d3

NIR · CRIRES+ J · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-d1f61926ee29

NIR · CRIRES+ J · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [10], [4] Ti-products-44ef55d79aae

NIR · CRIRES+ J · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-98bf194ecb60

NIR · CRIRES+ J · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-26b87bec7c59

NIR · CRIRES+ J · Ti I · reference lines · 1D LTE · [1], [2], [10], [4] Ti-products-fca24e3a90aa

NIR · CRIRES+ J · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [5] Ti-products-e151e5dc336b

NIR · CRIRES+ J · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-24a52395bbdf

NIR · CRIRES+ J · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [4] Ti-products-70c436249d65

NIR · CRIRES+ J · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-07cad2b0ccd2

NIR · CRIRES+ J · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-c83f37f35641

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D LTE · [1], [2], [9], [10], [4] Ti-products-c982225a9e5b

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [9], [10], [5] Ti-products-292ec65f11bf

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-bab26c42a214

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-4072d30f9dcb

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-ac2eed84f233

NIR · Kitt Peak (1984, molecfit) · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-64307ed5ab75

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D LTE · [1], [2], [9], [10], [4] Ti-products-09dcf4cc7d99

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [9], [10], [5] Ti-products-38791f738f64

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [4] Ti-products-41913ab66e30

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [9], [10], [4] Ti-products-5fd374e283c1

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [9], [10], [4], [6] Ti-products-cc758b3bbbab

NIR · Kitt Peak (1984, molecfit) · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [4], [6] Ti-products-b67f686e9433

VIS · HARPS · Ti I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [10], [8], [4] Ti-products-652a4d75c86a

VIS · HARPS · Ti I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [7], [10], [8], [5] Ti-products-82f4376a9fa4

VIS · HARPS · Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-66009f632479

VIS · HARPS · Ti I · Asplund et al. 2021 lines · 1D LTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-76be698fbb53

VIS · HARPS · Ti I · Asplund et al. 2021 lines · <3D> LTE (Gerber) · [1], [2], [7], [10], [8], [4], [6] Ti-products-3018829ae260

VIS · HARPS · Ti I · Asplund et al. 2021 lines · <3D> NLTE (Gerber) · [1], [2], [7], [10], [8], [4], [6] Ti-products-f12247355d78

VIS · HARPS · Ti I · Codex-graded lines · 1D LTE · [1], [2], [10], [8], [7], [4] Ti-products-bae71cfaabb3

VIS · HARPS · Ti I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [10], [8], [7], [5] Ti-products-aeb67743950d

VIS · HARPS · Ti I · Codex-graded lines · 1D NLTE (Gerber) · [1], [2], [10], [8], [7], [4] Ti-products-fa47dc01a3e5

VIS · HARPS · Ti I · Codex-graded lines · 1D LTE (Gerber) · [1], [2], [10], [8], [7], [4] Ti-products-b85e388e6622

VIS · HARPS · Ti I · Codex-graded lines · <3D> LTE (Gerber) · [1], [2], [10], [8], [7], [4], [6] Ti-products-20b4eb61acc2

VIS · HARPS · Ti I · Codex-graded lines · <3D> NLTE (Gerber) · [1], [2], [10], [8], [7], [4], [6] Ti-products-ec1b3fa73be0

VIS · HARPS · Ti I · reference lines · 1D LTE · [1], [2], [10], [9], [4] Ti-products-e899f67aa095

VIS · HARPS · Ti I · reference lines · 1D NLTE (Amarsi) · [1], [2], [10], [9], [5] Ti-products-fdbafd11ad48

VIS · HARPS · Ti I · reference lines · 1D NLTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-e217822e8473

VIS · HARPS · Ti I · reference lines · 1D LTE (Gerber) · [1], [2], [10], [9], [4] Ti-products-976826fbcb16

VIS · HARPS · Ti I · reference lines · <3D> LTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-054e7613991c

VIS · HARPS · Ti I · reference lines · <3D> NLTE (Gerber) · [1], [2], [10], [9], [4], [6] Ti-products-de52ddc63a01

VIS · Kitt Peak (Kurucz 2005) · Ti I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [7], [10], [8], [4] Ti-products-67b8068d849f

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VIS · Kitt Peak (1984, molecfit) · Ti I · Asplund et al. 2021 lines · 1D NLTE (Gerber) · [1], [2], [7], [10], [8], [4] Ti-products-7d2b12420af1

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